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EEL4712 Digital Design

Lecture 14 - VGA Timing & Pixel Coordinates

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Agenda

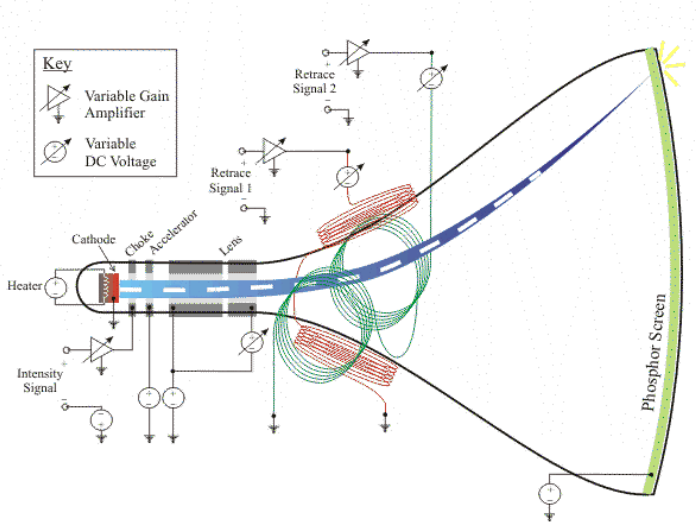
1. Motivation
2. VGA Overview
3. Horizontal and Vertical Timing Breakdown
4. Pixel Clock Generation
5. Horizontal and Vertical Counter Implementation
6. Synch Pulses and Pixels
7. Relationship to FSMD + Pittfalls
8. Summary

Motivation

- Interfaces follow protocols (VGA, UART, AXI, Avalon, etc)
- VGA is a real hardware protocol → precise timing, sync pulses
- Teaches real-time digital systems
- Reinforces counters, modular design, and FSMD concepts
- Preparation for more advanced video/graphics (beyond this course)

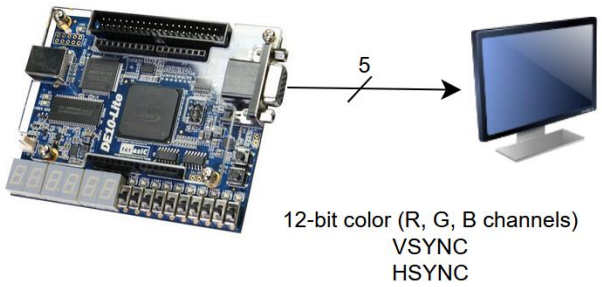
Video Graphics Array (VGA) Overview

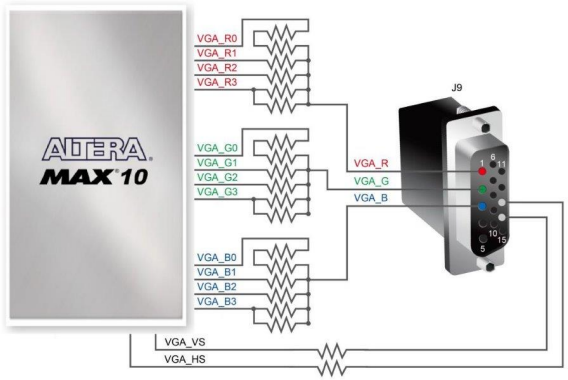
- Originally introduced by **IBM in 1987**
- Refers both to:
  - a **display standard** (resolution + timing), and
  - the familiar **15-pin D-sub (DE-15) connector**
  - Classic VGA resolution is **640 × 480 @ 60 Hz**

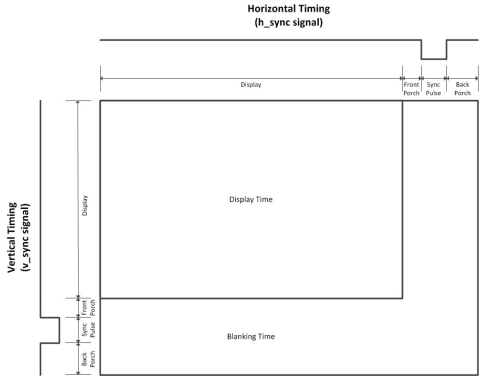


VGA Overview & Your FPGA

- VGA signal lines: R, G, B, HSYNC, VSYNC
- Resolution focus: 640×480 @ 60 Hz
- Pixel clock: 25.175 MHz (use 25 MHz on FPGA)
- Concept of visible region vs porch intervals
- Uses analog RGB signals plus sync signals (HSYNC, VSYNC)

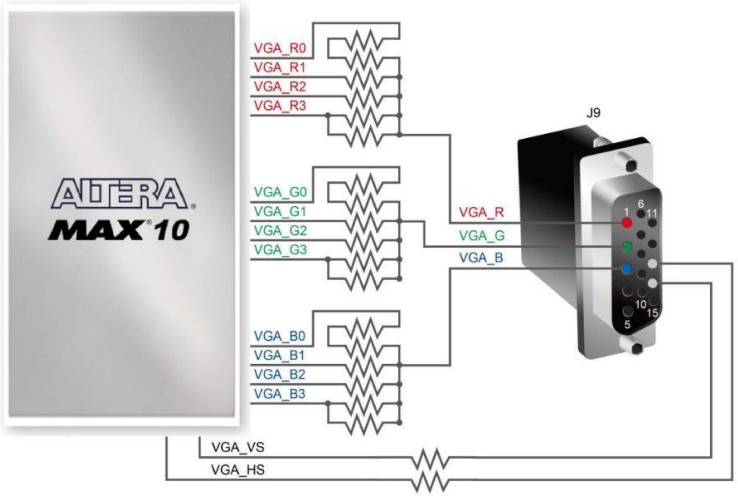






How Color Works

- Classic VGA:
- 3 analog signals:
  - Red
  - Green
  - Blue
  - Voltage level determines intensity.
  - On FPGA boards:
- Usually resistor ladder DAC
- 3–4 bits per color channel



Specification in Manual

- The specification on the FPGA board explains the expected format briefly
- Lecture slides will expand on that content

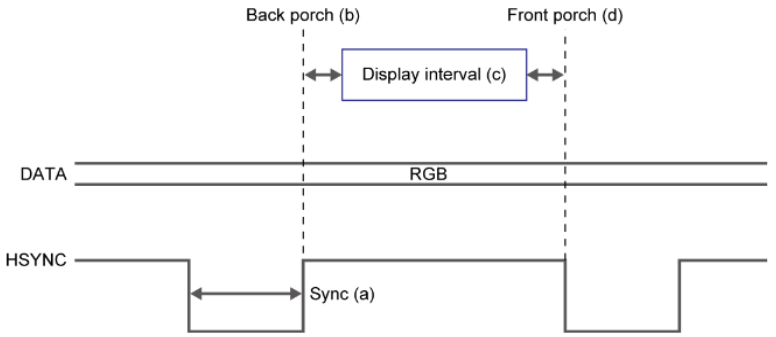


Figure 3-22 VGA horizontal timing specification

Table 3-9 VGA Horizontal Timing Specification

VGA mode		*Horizontal Timing Spec*				
Configuration	*Resolution(HxV)*	*a(pixel clock cycle)*	*b(pixel clock cycle)*	*c(pixel clock cycle)*	*d(pixel clock cycle)*	*Pixel clock(MHz)*
VGA(60Hz)	640x480	96	48	640	16	25

Table 3-10 VGA Vertical Timing Specification

VGA mode		*Vertical Timing Spec*				
Configuration	*Resolution(HxV)*	*a(lines)*	*b(lines)*	*c(lines)*	*d(lines)*	*Pixel clock(MHz)*
VGA(60Hz)	640x480	2	33	480	10	25

https://eel3701.ece.ufl.edu/assets/misc/DE10_Lite_User_Manual.pdf
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Specification in Manual

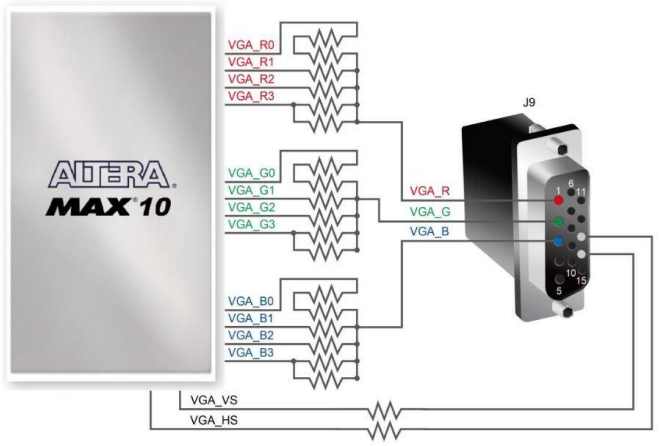


Table 3-11 Pin Assignment of VGA

Signal Name	FPGA Pin No.	Description	I/O Standard
VGA_R0	PIN_AA1	VGA Red[0]	3.3-V LVTTL
VGA_R1	PIN_V1	VGA Red[1]	3.3-V LVTTL
VGA_R2	PIN_Y2	VGA Red[2]	3.3-V LVTTL
VGA_R3	PIN_Y1	VGA Red[3]	3.3-V LVTTL
VGA_G0	PIN_W1	VGA Green[0]	3.3-V LVTTL
VGA_G1	PIN_T2	VGA Green[1]	3.3-V LVTTL
VGA_G2	PIN_R2	VGA Green[2]	3.3-V LVTTL
VGA_G3	PIN_R1	VGA Green[3]	3.3-V LVTTL
VGA_B0	PIN_P1	VGA Blue[0]	3.3-V LVTTL
VGA_B1	PIN_T1	VGA Blue[1]	3.3-V LVTTL
VGA_B2	PIN_P4	VGA Blue[2]	3.3-V LVTTL
VGA_B3	PIN_N2	VGA Blue[3]	3.3-V LVTTL
VGA_HS	PIN_N3	VGA Horizontal sync	3.3-V LVTTL
VGA_VS	PIN_N1	VGA Vertical sync	3.3-V LVTTL

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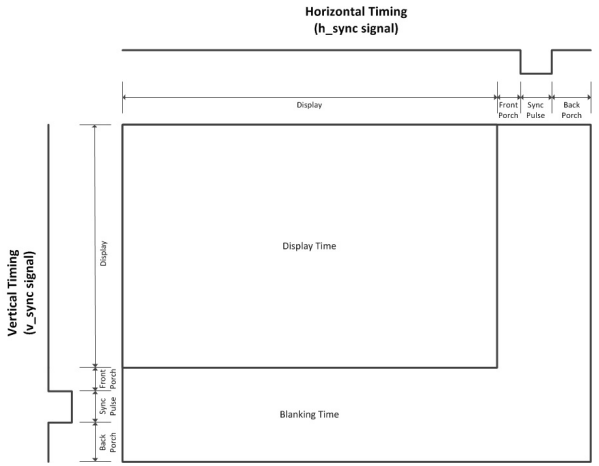
3. Pixel Clock Generation

- Desired Pixel clock: 25.175 MHz
- PLL or divider for 25 MHz
- Why clock accuracy matters
- Monitors have internal correction for 1-2% frequency errors

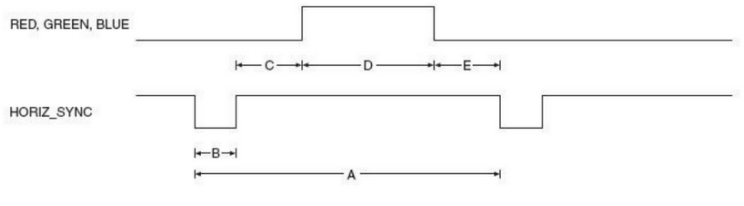
Timing and Conversion

640×480

- Convert time intervals to clock cycles
- T = 1/ 25MHz = 40 ns
- Cycles 25.17 MH / 40 ns => 629 cycles ~640

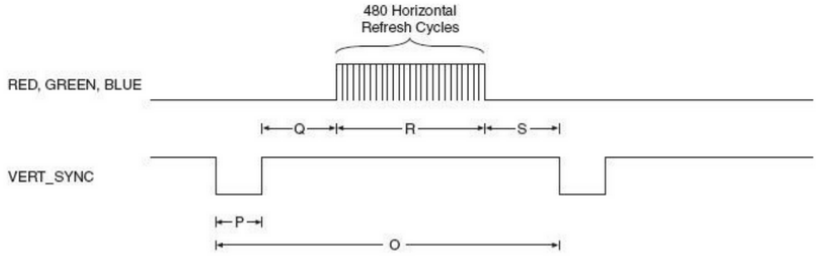


(The timing specifications are from a different board.)



Parameters	A	B	C	D	E
Time	31.77 µs	3.77 µs	1.89 µs	25.17 µs	0.94 µs

Figure 1. Horizontal Refresh Cycle.



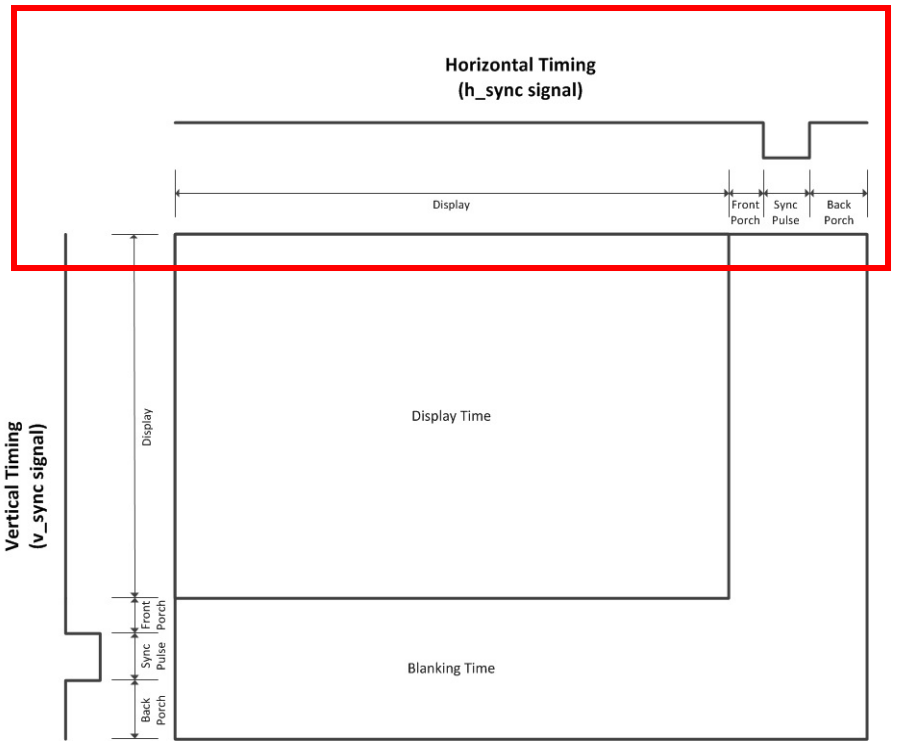
Parameters	O	P	Q	R	S
Time	16.6 ms	64 µs	1.02 ms	15.25 ms	0.35 ms

Figure 2. Vertical Refresh Cycle.

4. Horizontal Timing Breakdown

- Visible = 640 pixels
- Front porch = 16
- Sync pulse = 96
- Back porch = 48
- Total = **800 pixels per line**
- Waveform explanation:
- HSYNC low during sync pulse
- Blanking periods

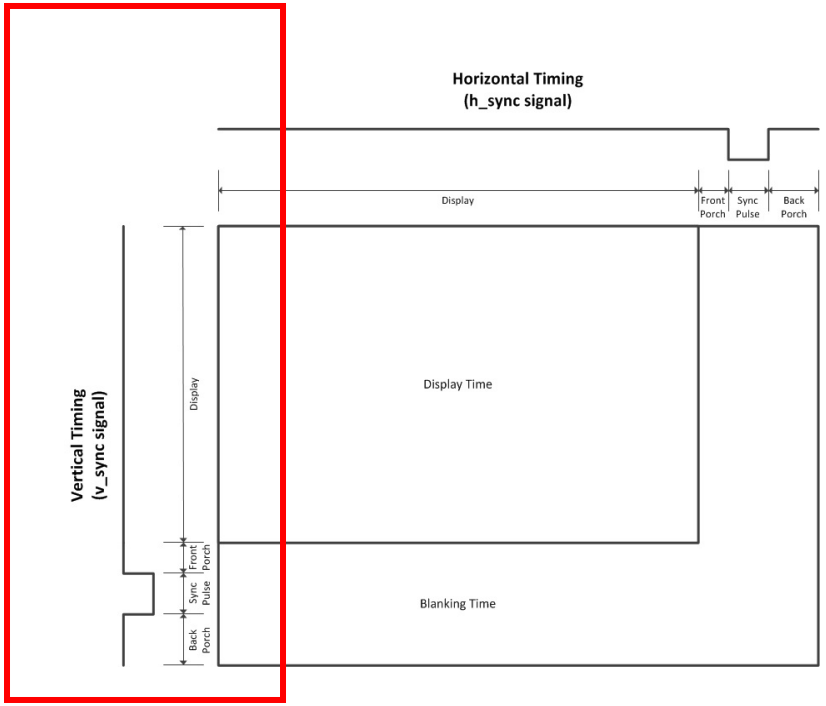
Why wrap at 699?




```
constant H_DISPLAY_END : integer := 639;
constant HSYNC_BEGIN : integer := 659;
constant H_VERT_INC : integer := 699;
constant HSYNC_END : integer := 755;
constant H_MAX : integer := 799;
```

5. Vertical Timing Breakdown

- Visible = 480 lines
- Front porch = 10
- Sync pulse = 2
- Back porch = 33
- Total = **525 lines per frame**
- VSYNC timing and polarity.




```
constant V_DISPLAY_END : integer := 479;
constant VSYNC_BEGIN   : integer := 493;
constant VSYNC_END     : integer := 494;
constant V_MAX         : integer := 524;
```

6. Horizontal Counter Implementation

- Counts from 0 → 799
- Wraps back to zero
- Drives pixel progression
- VHDL Sample:

```
signal hcount : integer range 0 to 799 := 0;

process(clk, rst)
begin
  if rst = '1' then
    hcount <= 0;
  elsif rising_edge(clk) then
    if hcount = 799 then
      hcount <= 0;
    else
      hcount <= hcount + 1;
    end if;
  end if;
end process;
```

7. Vertical Counter Implementation

- Increments when horizontal counter wraps
- Counts from 0 → 524

8. Generating Sync Pulses

- HSYNC
  - h >= 656 and h < 762 -> HSYNC = 0 else 1
- VSYNC
  - v >= 490 and v < 492 -> 0 else 1

9. Visible Region / Blanking

- display_on = (h < 640) and (v < 480)
- Only draw pixels when display_on = 1
- Else output black (0,0,0)

10. Pixel Coordinates

- From counters:
  - x = h
  - y = v
- Coordinates valid only during display region
- Useful for drawing shapes or backgrounds

11. Simple Pixel Generators

- Patterns using combinational logic:
  - Vertical color bars
  - Horizontal bars
  - Checkerboard
  - RGB gradients

Let's draw a block diagram with all the components

12. Relationship to FSMD

- FSMD triggers animation updates
- Example: move a square every N frames
- Switches (3 bit)
  - 0: Center
  - 1: Top left
  - 2: Top right
  - 3: Bottom left
  - 4: Bottom right

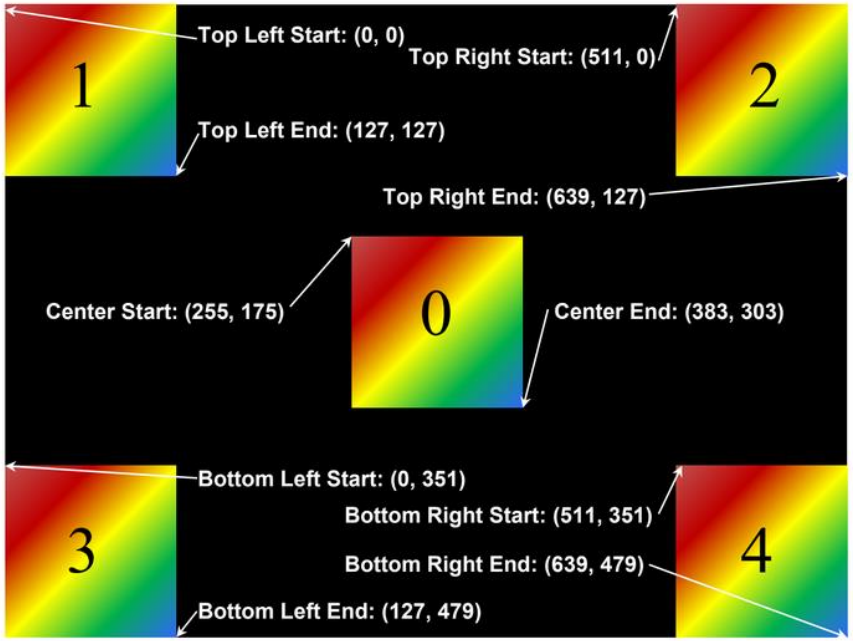


Figure 4. Display positions for different button combinations. Exactly one of these can be active at a time.

13. Common Mistakes

- Wrong sync polarity → monitor says “OUT OF RANGE”
- Off-by-one errors in timing constants
- Forgetting blanking: entire screen looks wrong

Summary

- Pixel clock
- Horizontal and vertical counters
- HSYNC, VSYNC generation
- Visible region
- Simple pixel generation using `(h, v)`